# Nature-Inspired Optimization

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**Date**: 05/08/2026

**Q.1 (1.00) -** Linear Relaxation consists of eliminating the integrality constraints of the variables. Regarding this technique, it is correct to state that:

a) ( ) The feasible region of the relaxed problem always disregards some feasible integer solutions of the original problem.
b) ( ) If the relaxed solution is fractional, the original problem has no feasible integer solution.
c) ( ) Rounding the solution obtained from linear relaxation always provides the optimal solution of the integer problem.
d) ( ) The feasible region of the relaxed problem is a subset of the feasible region of the original integer problem.
e) ( ) Rounding a relaxed solution may result in an infeasible solution for the original ILP problem.

**Q.2 (1.00) -** Consider the following optimization problem with an inequality constraint:

$$\min x^2 + 5y^2 + xy$$
$$\text{s.t.}$$
$$x^2 + y^2 \leq 1.$$

When solving this problem, an analyst must verify whether the inequality constraint is active or inactive at the minimum point. Based on optimization theory, mark the correct alternative about the impact of the constraint on the solution:

a) ( ) The objective function is non-convex, which prevents the use of the Karush-Kuhn-Tucker (KKT) conditions to verify whether the point is a global minimum.
b) ( ) The constraint makes the problem infeasible, since the Hessian matrix of the objective function has a negative determinant throughout the feasible region.
c) ( ) The constraint has a direct effect on the solution, because the global minimum of the original function lies exactly on the boundary of the feasible region.
d) ( ) The constraint has no effect on the minimum solution, because the critical point of the unconstrained problem already satisfies the inequality constraint.
e) ( ) According to the Lagrange Multipliers method, the constraint changes the minimum value to a negative value, making the problem unbounded.

**Q.3 (1.00) -** In the algebraic solution of unconstrained NLP problems, after finding a critical point through partial derivatives, it is necessary to classify this point. What is the usefulness of the Hessian Matrix and its leading principal minors in this process?
a) ( ) Transform the nonlinear function into a "Black Box" function.
b) ( ) Replace the need to compute the first-order derivatives of the objective function.
c) ( ) Compute the roots of the function through the Bisection method.
d) ( ) Determine the ideal learning rate for the Gradient method.
e) ( ) Determine the convexity or concavity of the function to identify whether the critical point is a maximum, minimum, or saddle point.

**Q.4 (1.00) -** The classification of problems into complexity classes is crucial to determine the solution strategy. Regarding the P and NP classes, according to the sources, mark the correct alternative:

a) ( ) If a problem is NP-complete, then it does not have a polynomial verifier.
b) ( ) All combinatorial optimization problems necessarily belong to class P.
c) ( ) Problems in class NP are those that cannot be solved in polynomial time.
d) ( ) Class P comprises problems for which there exists a polynomial-time algorithm that finds the solution.
e) ( ) A problem is in P if it is possible to verify a proposed solution in polynomial time, even if one does not know how to find it in polynomial time.

**Q.5 (1.00) -** Suppose a computer assembly company manufactures three basic types of devices: desktops, laptops, and tablets. The company's production takes place in its two factories, one in Manaus and the other in the São Paulo. According to the last sale, the company needs to deliver 16,000 desktops, 6,000 laptops, and 28,000 tablets. The Manaus factory has a daily production cost of R$150,000.00 for the production of 8,000 desktops, 1,000 laptops, and 2,000 tablets. The daily production cost of the factory in São Paulo is R$210,000.00 for the production of 2,000 desktops, 1,000 laptops, and 7,000 tablets.

a) (   ) Min $Z = 150{,}000x + 210{,}000y$
s.t.
$8x+2y = 16;$
$x+y = 6;$
$2x+7y \geq 28$

b) (   ) Min $Z = 150{,}000x + 210{,}000y$
s.t.
$8x+2y \leq 16;$
$x+y \leq 6;$
$2x+7y \leq 28$

c) (   ) Min $Z = 150{,}000x + 210{,}000y$
s.t.
$8x+2y \geq 16;$
$x+y \geq 6;$
$2x+7y \geq 28$

d) (   ) Min $Z = 150{,}000x + 210{,}000y$
s.t.
$8x-2y \leq 16;$
$x+y \leq 6;$
$2x-7y \leq 28$

e) (   ) Min $Z = 150{,}000x + 210{,}000y$
s.t.
$8x-2y \leq 16;$
$x-y \leq 6;$
$2x-7y \leq 28$

**Q.6 (1.00) -** In the process of engineering and problem solving, abstraction is defined as the mental operation of considering only some characteristics of a phenomenon. When building a model to represent reality, the professional must establish a fundamental trade-off. What is this trade-off?

a) (   ) Between the complexity of the model and its ability to represent the problem of interest.
b) (   ) Between the number of decision variables and the number of equality constraints of the system.
c) (   ) Between the use of “White Box” models and “Gray Box” models.
d) (   ) Between the cost of the computational solver and the precision of the hardware used.
e) (   ) Between the use of discrete variables and the use of continuous variables.

**Q.7 (1.00) -** Regarding the relationship between Integer Programming (IP) problems and the complexity classes P and NP, mark the correct alternative:

a) (   ) The Branch-and-Bound method is considered a polynomial-time algorithm, which proves that Integer Linear Programming and Linear Programming have the same computational complexity.
b) (   ) Some integer programming problems, such as the Shortest Path and Assignment problems, belong to class P and linear relaxation often provides solutions that already satisfy integrality conditions.
c) (   ) Rounding the solution of a linear relaxation guarantees converting an NP-hard problem into a P-class problem, always maintaining the optimality of the integer solution.
d) (   ) All problems that use discrete decision variables (integer or binary) are necessarily NP-hard, always requiring the use of metaheuristics to find feasible solutions.
e) (   ) The Knapsack Problem is considered P because no algorithm is known that finds the optimal solution in polynomial time for all instances.

**Q.8 (1.00) -** Consider the following code snippet written in Python using the Pyomo library to model an optimization problem. Based on the code, mark the alternative that correctly presents the corresponding mathematical model and its classification:

```
from pyomo.environ import *
model = ConcreteModel()
model.x = Var(within=NonNegativeIntegers)
model.y = Var(within=NonNegativeIntegers)
model.obj = Objective(expr=50 * model.x + 80 * model.y, sense=maximize)
model.con1 = Constraint(expr=2 * model.x + 4 * model.y <= 20)
model.con2 = Constraint(expr=3 * model.x + 2 * model.y <= 18)
```

a) ( ) max $50x + 80y$
s.t.
$2x+4y \leq 20$;
$3x+2y \leq 18$;
$x,y \in Z+$
**Classification**: Integer Linear Programming (ILP)

b) ( ) max $50x + 80y$
s.t.
$2x+4y \leq 20$;
$3x+2y \leq 18$;
$x,y \in \{0,1\}$
**Classification**: Binary Programming (BIP)

c) ( ) min $50x + 80y$
s.t.
$2x+4y \leq 20$;
$3x+2y \leq 18$;
$x,y \geq 0$
**Classification**: Linear Programming (LP)

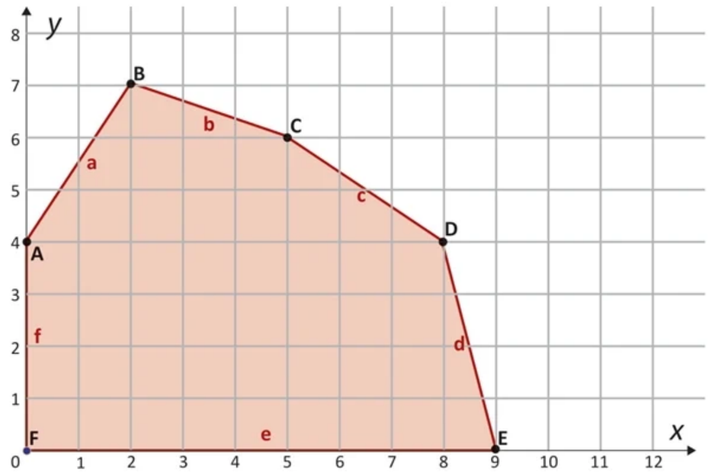
d) ( ) max $50x^2 + 80y^2$
s.t.
$2x+4y = 20$;
$3x+2y = 18$;
$x,y \geq 0$
**Classification**: Nonlinear Programming (NLP)

e) ( ) min $50x + 80y$
s.t.
$2x+4y \geq 20$;
$3x+2y \geq 18$;
$x,y \in R$
**Classification**: Linear Programming (LP)

**Q.9 (1.00) -** To use numerical methods such as Simplex, it is necessary to transform the Linear Programming model into the standard form, where all constraints are equalities. How can an inequality constraint of the form $ax + by \geq c$ be converted into an equality?

a) ( ) By substituting the original variables with free variables.
b) ( ) By multiplying all coefficients by −1 and changing the "≥" sign to "=".
c) ( ) By multiplying the left-hand side of the equation by a surplus variable.
d) ( ) By subtracting a surplus variable from the left-hand side of the inequality.
e) ( ) By dividing the left-hand side of the equation by a surplus variable.

**Q.10 (1.00) -** In a computational system, performance is evaluated for the use of two types of secondary memory through linear programming. In the presented graph, each axis represents the quantity of memory slots of each type and the colored region continuously represents the set of feasible solutions (feasible region). The edges of the feasible region were defined by the various constraints observed for the two types of memory. Based on the graph, the maximum system performance, represented by the objective function 30x + 10y, is:




a) ( ) 280
b) ( ) 200
c) ( ) 270
d) ( ) 180
e) ( ) 130